Logistic Regression and Classification

ANSWER KEY

Section 1: The Odds Ratio and Logit Formulation

1 If a classifier predicts probability $p = 0.80$ for disease occurrence, what are the odds of having the disease?

a. 4.0 ✓

b. 0.25

c. 0.80

d. 5.0

$\text{Odds} = \frac{p}{1-p} = \frac{0.80}{0.20} = 4.0$ (or 4 to 1).

2 Compute the log-odds (logit) value when probability $p = 0.50$.

a. 0.0 ✓

b. 1.0

c. 0.5

d. -1.0

$\text{logit}(0.5) = \ln\left(\frac{0.5}{0.5}\right) = \ln(1) = 0.0$.

3 If a logistic model has equation $\ln\left(\frac{p}{1-p}\right) = 2.0$, find the predicted probability p.

a. $\frac{e^2}{1+e^2} \approx 0.88$ ✓

b. $\frac{1}{2} = 0.50$

c. $\frac{2}{3} \approx 0.67$

d. 0.95

Inverting logit: $p = \sigma(2.0) = \frac{e^2}{1+e^2} = \frac{7.389}{8.389} \approx 0.88$.

Section 2: Binary Cross-Entropy Loss

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4 Calculate the binary cross-entropy loss $L = -[y \ln \hat{y} + (1-y)\ln(1-\hat{y})]$ for a sample where true label $y = 1$ and predicted probability $\hat{y} = 0.90$.

a. $-\ln(0.90) \approx 0.105$ ✓

b. $-\ln(0.10) \approx 2.30$

c. 0.90

d. 0.0

When $y = 1$: $L = -1\ln(0.90) - 0 = -\ln(0.90) \approx 0.105$.

5 What is the derivative of the binary cross-entropy loss with respect to the pre-activation logit z, where $\hat{y} = \sigma(z)$?

a. $\hat{y} - y$ ✓

b. $\hat{y}(1-\hat{y})$

c. $y - \hat{y}$

d. $\frac{\hat{y}-y}{\hat{y}(1-\hat{y})}$

Remarkably clean: $\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}}\frac{\partial \hat{y}}{\partial z} = -\left(\frac{y}{\hat{y}} - \frac{1-y}{1-\hat{y}}\right)\hat{y}(1-\hat{y}) = \hat{y} - y$.

6 What happens to the cross-entropy loss as $\hat{y} \to 0$ when true label is $y = 1$?

a. **It approaches $+\infty$, heavily penalizing confident wrong predictions** ✓

b. It approaches zero

c. It oscillates around 1

d. It equals the pre-activation logit

$L = -\ln(\hat{y})$. As $\hat{y} \to 0$, $-\ln(\hat{y}) \to +\infty$.

Section 3: The Confusion Matrix and Core Metrics

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7 A diagnostic model reports $TP = 70, TN = 20, FP = 5, FN = 5$. Compute the Accuracy.

a. 90% ✓

b. 85%

c. 93.3%

d. 75%

$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} = \frac{70 + 20}{100} = 90\%.$

8 Using the same counts ($TP = 70, FP = 5, FN = 5$), calculate the model's Precision.

a. $\frac{70}{75} \approx 93.3\%$ ✓

b. $\frac{70}{70 + 5} = 90\%$

c. $\frac{70}{100} = 70\%$

d. $\frac{75}{100} = 75\%$

$\text{Precision} = \frac{TP}{TP + FP} = \frac{70}{75} \approx 93.3\%.$

9 Calculate the Recall (Sensitivity) for the model above.

a. $\frac{70}{75} \approx 93.3\%$ ✓

b. $\frac{70}{90} \approx 77.8\%$

c. $\frac{5}{75} \approx 6.7\%$

d. 85%

$\text{Recall} = \frac{TP}{TP + FN} = \frac{70}{70 + 5} = \frac{70}{75} \approx 93.3\%.$

Section 4: The F_1-Score, Thresholding and ROC-AUC

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10 Compute the F_1-score for a model having Precision $= 0.80$ and Recall $= 0.60$.

a. $\frac{2(0.8)(0.6)}{0.8+0.6} \approx 0.686$ ✓

b. 0.700

c. 0.480

d. 0.750

F_1 is the harmonic mean: $2\frac{P \cdot R}{P+R} = \frac{2(0.48)}{1.40} = \frac{0.96}{1.40} \approx 0.686$.

11 If you lower the classification probability threshold from 0.5 to 0.2, what is the typical effect on Precision and Recall?

a. **Recall increases (fewer false negatives), while Precision typically decreases (more false alarms)** ✓

b. Precision increases while Recall decreases

c. Both metrics increase simultaneously

d. Neither metric changes

A lower threshold flags more samples positive: catches more real cases (high Recall), but flags more false alarms (lower Precision).

12 What does an Area Under the ROC Curve (ROC-AUC) of 0.50 indicate about a classifier?

a. **The model has no discriminative ability (equivalent to random guessing)** ✓

b. The model is a perfect classifier

c. The model achieves 50% precision at all thresholds

d. The model is inverted

The diagonal line from $(0, 0)$ to $(1, 1)$ represents random guessing and has area under the curve $= 0.50$.

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